

STAT 380 Data Science Through Statistical Reasoning and Computation – Mini-Project 2

Due: By 11:59 PM on Thursday, April 3, 2025

Instructions

The purpose of this Mini-Project is to have you continue your exploration of a dataset. Some of the tasks required to do this project have not been covered in STAT 380. This is typical of a project in which you have some knowledge, but you need to develop new skills as well.

Data

The dataset CODGames2_mp.xlsx should be downloaded from Canvas. The dataset contains information about the results of an online, first-person shooter video game. In the game, the player is part of a team trying to eliminate enemy combatants and/or achieve objectives. Each row represents the results of a single online match. At this time, no description of the variables is available.

Submission

Each student should submit a report and code that addresses the tasks listed below. (The easiest way to create the report is to use a Markdown document so that your code, plots, and explanations are next to each other as you do in the HW.) **Be sure to submit both your .rmd file and your .html file.**

This assignment is meant to be more formal than a typical HW assignment. A portion of the grade (approximately 4-5 points out of 30 points) will be assigned regarding the neatness and quality of your work. Be sure to proofread to correct typos, fix headings that are not displaying properly, correct hanging sentences (sentences/paragraphs that start immediately to the right of a plot but conclude below the plot), ensure plots are readable and use customized axis titles rather than raw variable names, be sure that discussions are not R comments placed in R chunks, and make an effort to write thoughtful discussions.

For this assignment, you have the option to work alone or with a small group. Group size is limited to at most 3 members. If you know nothing about these types of video games, it may be helpful to work with someone that does. This is a good opportunity to try out teammates for the final project.

Each student should submit the report. If you work with other students, include the names of any students that you worked with in your report and a discussion of each group member's contributions. (All group members should submit the same report.) Although students like a divide-and-conquer approach to reduce the amount of work, all group members are expected to read and contribute to the work done by others.

Tasks

1. Explain why you should avoid using the function na.omit() to remove observations (rows) with missing data for the CODGames2_mp dataset. NOTE: If you want to write a great answer, use the function that we wrote in the notes for counting the number of NA's and apply that function to each row of the dataset. (In the notes we applied the function to each column instead of each row.)

2. In Mini-Project 1, you were told to “Begin” exploring the dataset by exploring 5 variables of your choice. Continue that exploration. Explore 5 more variables of your choice. (You should explore at least one categorical and at least one quantitative variable.) Perform a univariate exploration of each variable. In other words, do not look at the relationship between pairs of variables. Instead, just explore each variable by itself.

- For each, you should identify the variable, the type of variable (categorical, quantitative, etc.), information about the types of values the variable can take, and the amount of missing data within the variable.
- For each variable, report some summary statistics and make a visualization.
 - If a variable is categorical, make a table of possible values and include counts or proportions. Then, make an appropriate plot to show the distribution of values.
 - If a variable is quantitative, include some basic summary stats like the mean, median, standard deviation, min, max, etc. and make an appropriate plot to show the distribution of values.
- For each variable, you should write a few sentences to describe what you have learned about the variable. Incorporate the summary statistics and comment on the amount of missingness into your discussion. Remember, the goal is not to make the plot/calculate the summaries. The goal is to use plots and summaries to gain knowledge from the data.
- Place each discussion next to the relevant plot/summary statistics rather than writing code for all 5 variables first followed by all discussion. If you the discussions are not near the plots, it makes it difficult for your audience to follow your explanations without excessive scrolling.

3. This exercise relates to the column titled ‘Result’. The ‘Result’ variable contains the scores of each online match. Each match involves two teams of players. Teams earn points for completing certain tasks that depend on the game mode. The match ends when either a) the time limit expires or b) one of the teams reaches the score limit (varies based on game mode). At the end of the match, the team with the higher score is considered the winner. Some notes:

- The Result is reported as a character string with the team scores separated by a hyphen.
- The number to the left of the hyphen is the score for the player’s team.
- The number to the right of the hyphen is the score for the other team.
- Examples:
 - Row 1: “100-97” means that the player’s team won by a score of 100 to 97
 - Row 2: “76-89” means that the player’s team lost by a score of 89 to 76
 - Row 36: “200-200” means that the teams played to a draw (i.e., tied 200 to 200)
- The match result is either a win, loss, or draw (occurs when teams finish with the same number of points)

Your task is to write code to calculate the proportion of games the player won, the proportion of games the player lost, and the proportion of games that resulted in a draw. To full credit, the code must calculate the proportions in a general way instead of a way that is specific to your dataset (e.g., do not divide by typing /211 (since 211 is the exact number of rows). Write code that would work regardless of whether the dataset has 211 rows or 2000 rows. NOTE: While you may create a plot that shows the proportions, I still want to see the actual numeric values of the proportions.

HINTS:

- We have not covered this topic in our class. So, you will either rely on knowledge from STAT 184 (or another course) or web search for a solution. This is the nature of real work. You will encounter tasks you do not know how to handle. Searching online resources is the key.
- If you have not worked with character strings before, you will want to find a way to split the string in the 'Result' variable into 2 scores (one for the player's team, one for the other team), convert the values from character to numeric, make a comparison, and based on the comparison decide if the match was a win, loss, or draw.
- If you/your team has worked on this for 1 hour, send me a list of 5 google searches (or 5 pieces of code) you have tried unsuccessfully, and I will give you a better hint.

4. WE WILL DISCUSS THIS IDEA IN CLASS ON MONDAY, MARCH 31. (Asking and answering an inference question that can be answered using **logistic regression**.) For this task, you will write your own inference question by building a model in which the response variable is whether the player's team won the game. Treat a game in which the teams tied as not winning. Then, you will answer build the model and answer the question using the model. Some notes:

- Inference questions are often about exploring the relationship between variables (See HW5, Question 5 for an example of an inference question in linear regression).
- Inference questions are different than questions about the accuracy of predictions. So, this should **not** be a question that you answer by assessing validation set performance.
- When interpreting coefficients in logistic regression, be sure that your interpretation is about the odds rather than the log odds.

NOTE: Part of your grade for this question will be determined based on whether your question is a valid question about inference and whether your analysis allows you to answer the question.